

多維度連續有界資料的強韌概似分析

—以人體體脂百分比資料為例

摘要

有界連續型資料於醫學研究領域中十分常見，尤其比率資料常作為醫療判斷或參照的指標，例如本文實例－人體體脂百分比資料。該資料收集了每個人手臂、軀幹、腹部、臀部及腿部五個不同部位的體脂率，由於這些觀察值來自同一個人所提供，因此資料往往會具有相關性，此時便可考慮使用多元廣義線性混和模型 (multivariate generalized linear mixed models) 進行描述及分析。然而，當資料維度增加時，計算最大概似估計量(maximum likelihood estimation)常會面臨高維度積分的問題，需仰賴耗時的數值積分或其他積分近似方法進行計算，並且在未知資料真實分配的情況，若分配模型假設錯誤，也可能得到不合適或不正確的結果。

本文利用強韌化多元負二項分配 (multivariate negative binomial distribution) 之概似函數分析多維連續有界資料。除了在錯誤模型假設下，仍能對感興趣之參數提供具一致性的估計量外，計算過程也不需對模型進行積分。而本文模擬研究與實例分析中所呈現的強韌華德檢定統計量 (robust wald statistics)、強韌分數檢定統計量 (robust score statistics) 及強韌概似比檢定統計量 (robust likelihood ratio statistics)，說明了強韌概似方法方法在資料真實分配未知的狀況下，仍可做出正確的統計推論。

關鍵字：強韌概似函數、多元負二項分配、廣義線性混和模型、多維度連續有界資料

Robust Likelihood approach for multivariate continuous bounded data — taking the body fat percentage data for example

Abstract

In clinical studies and biomedical research, it is common to encounter continuous bounded data, such as body fat percentage. This paper focuses on a dataset that includes body fat percentages measured at five regions of each body: arms, legs, trunk, android, and gynoid. Since these five observations come from the same individual, correlations exist in different responses. Some researchers choose multivariate generalized linear mixed models (MGLMMs) to model this type of data. However, when dealing with high-dimensional data, estimating the maximum likelihood often faces challenges due to high-dimensional integration. Furthermore, if the model is misspecified, the analysis may yield incorrect results.

This paper applies a robust multivariate negative binomial likelihood function to analyze multivariate continuous bounded data. In addition to consistent estimates for the parameters of interest, the adjusted likelihood function enables obtaining correct asymptotic variance estimates. Moreover, the robust Wald statistics, robust score statistics, and robust likelihood ratio statistics presented in this paper show that the robust likelihood approach can always make correct statistical inferences even if the true underlying distribution is unknown.